

Banker's Algorithm

3. a) Five processes and 3 resources types A, B, C and D. Below is the snapshot of the state as:

Process	Maximum				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	6	0	1	2	4	0	0	1	3	2	1	1
P ₁	2	7	5	0	1	1	0	0				
P ₂	2	3	5	6	1	2	5	4				
P ₃	1	6	5	3	0	6	3	3				
P ₄	1	6	5	6	0	2	1	2				

sq Is this a safe state? If yes, what is the sequence?

sq Here,

$$\text{Need} = \text{Maximum} - \text{Allocation.}$$

Process	Maximum				Allocation				Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	6	0	1	2	4	0	0	1	2	0	1	1	3	2	1	1
P ₁	2	7	5	0	1	1	0	0	1	6	5	0				
P ₂	2	3	5	6	1	2	5	4	1	1	0	2				
P ₃	1	6	5	3	0	6	3	3	1	0	0	0				
P ₄	1	6	5	6	0	2	1	2	1	4	4	4				

i> The content of A matrix is:

Process	Need			
	A	B	C	D
P ₀	2	0	1	1
P ₁	1	6	5	0
P ₂	1	1	0	2
P ₃	1	0	2	0
P ₄	1	4	4	4

ii) We have, $A = (3, 2, 1, 1)$

Step 1: With the current available resources, $A = (3, 2, 1, 1)$, P_0 can be executed.

Since the need of $P_0 \leq A$ i.e. $(2, 0, 1, 1) \leq (3, 2, 1, 1)$

Now the total current available resources A

= Previous resources + Allocated by P_0

$$= (3, 2, 1, 1) + (4, 0, 0, 1) = (7, 2, 1, 2)$$

Step 2: With the current available resources, $A = (7, 2, 1, 2)$

P_2 can be executed.

Since the need of $P_2 \leq A$ i.e. $(1, 1, 0, 2) \leq (7, 2, 1, 2)$

Now the total current available resources A

= Previous resources + Allocated by P_2

$$= (7, 2, 1, 2) + (1, 2, 5, 4) = (8, 4, 6, 6)$$

Step 3: With the current available resources, $A = (8, 4, 6, 6)$

P_3 can be executed.

Since need of $P_3 \leq A$ i.e. $(1, 0, 2, 0) \leq (8, 4, 6, 6)$

Now the total current available resources A

= Previous free + Allocated by P_3

$$= (8, 4, 6, 6) + (0, 6, 3, 3) = (8, 10, 9, 9)$$

Step 4: With the current available resources, $A = (8, 10, 9, 9)$

P_1 can be executed.

Since need of $P_1 \leq A$ i.e. $(1, 6, 5, 0) \leq (8, 10, 9, 9)$

Now the total current available resource A

= Previous free + Allocated by P_1

$$= (8, 10, 9, 9) + (1, 1, 0, 0) = (9, 11, 9, 9)$$

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Step 5: With the current available resource $A = (9, 11, 9, 9)$. P_4 can be executed,
 Since need of $P_4 \leq A$ i.e. $(1, 4, 4, 4) \leq (9, 11, 9, 9)$
 Now the total current available resources A
 $=$ previous free + Allocated by P_4
 $= (9, 11, 9, 9) + (0, 2, 1, 2) = (9, 13, 10, 11)$.

Since all the process are executed. Hence they are in safe state. safe sequence:

$P_0 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1 \rightarrow P_4$. Ans

2. Consider a system 5 processes: P_0, P_1, P_2, P_3 & P_4 and three resources type: A, B, C. A has 10 instances, B has 5 instances and C has 7 instances. Suppose at time interval to following snapshot of the system has been taken. Check whether the process is safe or not using Banker's algorithm. If safe, write the safe sequence.

Process	Allocation			Maximum			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	7	5	3	3	2	2
P_1	2	0	0	3	2	2			
P_2	3	0	2	9	0	2			
P_3	2	1	1	2	2	2			
P_4	0	0	4	4	3	3			

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Here,

$$\text{Need} = \text{Maximum} - \text{Allocation}$$

Process	Allocation			Maximum			Available			Need		
	A	B	C	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3	3	3	2	7	4	3
P ₁	2	0	0	3	2	2				1	2	2
P ₂	3	0	2	9	0	2				6	0	0
P ₃	2	1	1	2	2	2				0	1	1
P ₄	0	0	2	4	3	3				4	3	1

i) The content of A matrix:

Process	A	B	C
P ₀	7	4	3
P ₁	1	2	2
P ₂	6	0	0
P ₃	0	1	1
P ₄	4	3	1

ii) We have, $A = (3, 3, 2)$

Step 1: With the current available resources $A = (3, 3, 2)$.

P₁ can be executed.

Since need of P₁ $\leq A$ i.e. $(1, 2, 2) \leq (3, 3, 2)$.

Now the current available (total) resources A
= Previous free + Allocated by P₁

$$= (3, 3, 2) + (2, 0, 0) = (5, 3, 2) = (5, 3, 2)$$

Step 2: With the current available resources, $A = (5, 3, 2)$.

P_3 can be executed.

Since the need of $P_3 \leq A$ i.e. $(0, 1, 1) \leq (5, 3, 2)$.

~~Now~~ Now the total current available resources

$A =$ previous free + Allocated by P_3

$$= (5, 3, 2) + (2, 1, 1) = (7, 4, 3)$$

Step 3: With the current available resources, $A = (7, 4, 3)$.

P_4 can be executed.

Since need of $P_4 \leq A$ i.e. $(4, 3, 1) \leq (7, 4, 3)$.

Now the total current available resources A

$=$ previous free + Allocated by P_4

$$= (7, 4, 3) + (0, 0, 4) = (7, 4, 7)$$

Step 4: With the current available resources, $A = (7, 4, 7)$

P_0 can be executed.

Since need of $P_0 \leq A$ i.e. $(7, 4, 3) \leq (7, 4, 7)$.

Now the total current available resources A

$=$ previous free + Allocated by P_0

$$= (7, 4, 7) + (0, 1, 0) = (7, 5, 7)$$

Step 5: With the current available resources, $A = (7, 5, 7)$

P_2 can be executed.

Since need of $P_2 \leq A$ i.e. $(7, 5, 7) \leq (7, 5, 7)$ ~~(6, 0, 0) $\leq$ (7, 5, 7)~~

Now the total current available resources A

$=$ previous free + Allocated by P_2

$$= (7, 5, 7) + (3, 0, 2) = (10, 5, 9)$$

Since all processes are executed, so the processes are in safe state. Safe Sequence: $P_1 \rightarrow P_3 \rightarrow P_4 \rightarrow P_0 \rightarrow P_2$.

2020 - Fall

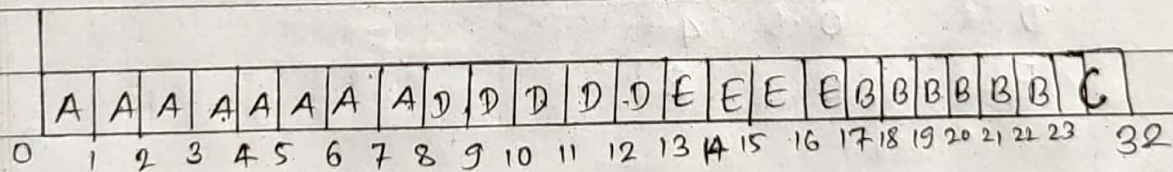
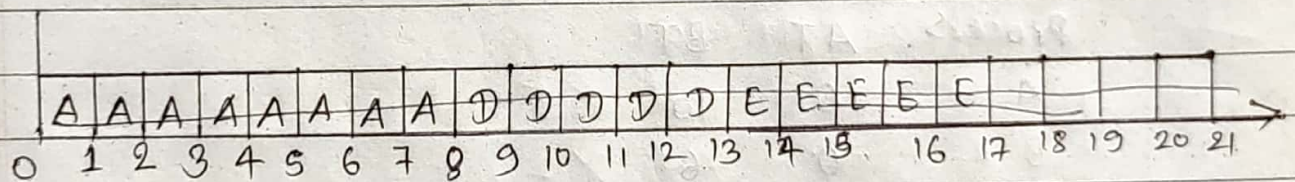
4a) Given the following set of information, what is the AWT and ATAT using SJF (Preemptive), FCFS, RR (Quantum=3) and HRRN.

Process	AT	BT	
A	0	8	7 6 5 4 3 2 1 0
B	2	6	5 4 3 2 1 0
C	4	9 9	
D	7	5	4 3 2 1 0
E	9	4	3, 2, 1, 0

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i) SJF (Preemptive) / (SRTF):

⇒ * Gantt chart:

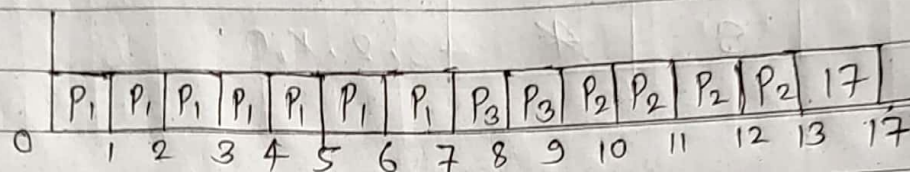


* SJF (Preemptive) / SRTF

2021 - Fall

①	Process	AT	BT
	P ₁	0.0	7.0 8.0 3.0 2.0 1.0
	P ₂	3.0	4.0 3.0 1.0
	P ₃	5.0	2.0 1.0
	P ₄	6.0	4

⇒ Gantt chart:

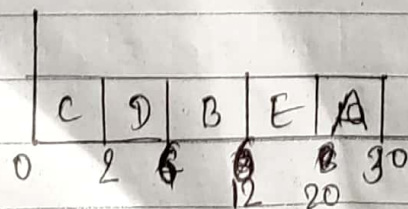


2021 - Spring:

① SJF (Non-preemptive)

Process	AT	BT
A	0	10
B	0	6
C	0	2
D	0	4
E	0	8

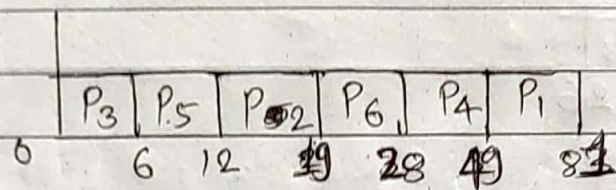
⇒ Gantt chart:



2022-Fall

SJF (Preemptive):

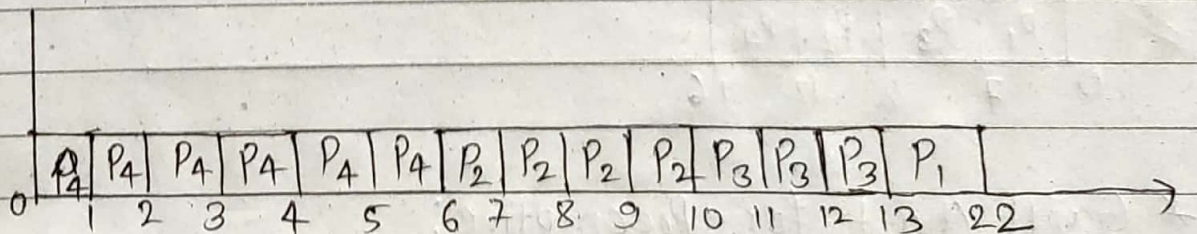
Process	AT	BT
P ₁	0	32
P ₂	0	7
P ₃	0	6
P ₄	0	21
P ₅	0	6
P ₆	0	9



2023 Spring

Process	AT	BT
P ₁	0	9
P ₂	4	4 3 2 1 0
P ₃	10	3 2 1 0
P ₄	0	6 5 4 3 2 1 0

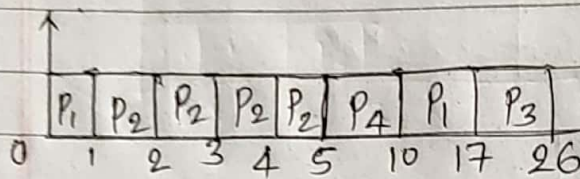
Gantt chart:



Q Notes SJF (Preemptive)

Process	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

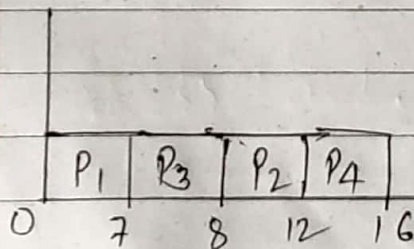
Gantt chart:



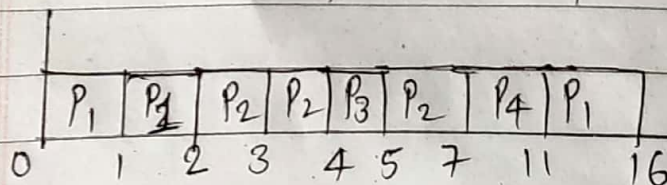
Q SJF (Non-Preemptive)

Process	AT	BT
P ₁	0.0	7
P ₂	2.0	4
P ₃	4.0	0
P ₄	5.0	4

* Gantt chart



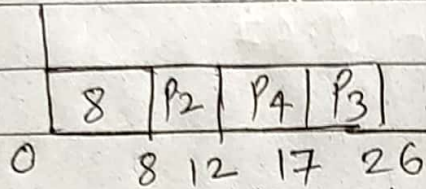
* Gantt Chart for Preemptive:



SJF (Non-preemptive),

Q

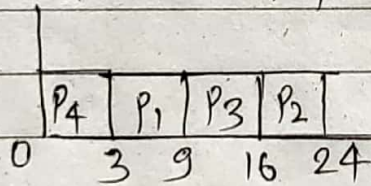
Process	AT	BT
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5



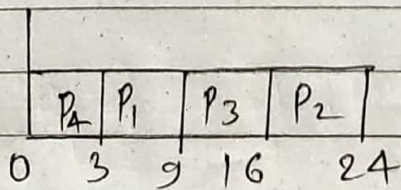
Q: SJF:

Process	BT
P ₁	6
P ₂	8
P ₃	7
P ₄	3

Gantt chart for Non-preemptive



Gantt chart for Preemptive:

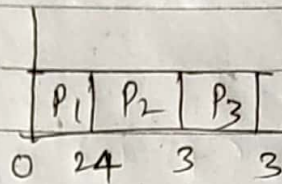


FCFS

Q:

Process	Burst Time
P ₁	24
P ₂	3
P ₃	3

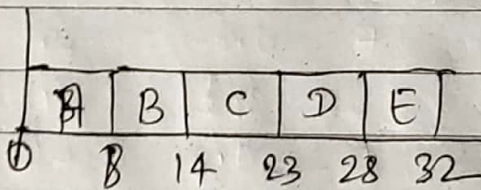
* Gantt Chart:



Q:

Process	AT	BT
A	0	8
B	2	6
C	4	9
D	7	5
E	9	4

2020
- Fall



Q: Process

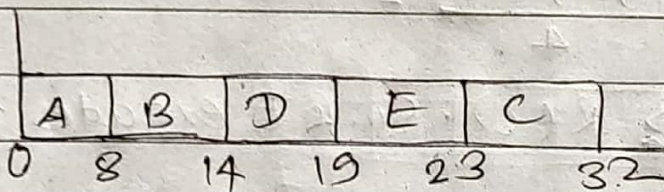
HRRN

2020-Fall

Q: Process:

	AT	BT
A	0	8
B	2	6
C	4	9
D	7	5
E	9	4

* Gantt chart:



$$RR(\text{Process}) = \frac{\text{Time Now} - AT + BT}{BT}$$

At time = 8:

$$RR(B) = \frac{8 - 2 + 6}{6} = 2$$

$$RR(C) = \frac{8 - 4 + 9}{9} = 1.44$$

$$RR(D) = \frac{8 - 7 + 5}{5} = 1.2$$

$\therefore RR(B) \geq RR(C) \geq RR(D)$, so B executes.

At time = 14:

$$RR(C) = \frac{14 - 4 + 9}{9} = 2.11$$

$$RR(D) = \frac{14 - 7 + 5}{5} = 2.4$$

$$RR(E) = \frac{14 - 9 + 4}{4} = 2.25$$

$\therefore RR(D) \geq RR(E) \geq RR(C)$. So D executes.

At time = 19:

$$RR(C) = \frac{19 - 4 + 9}{9} = 2.67$$

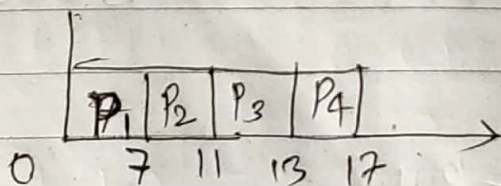
$$RR(E) = \frac{19 - 9 + 4}{4} = 3.5$$

$\therefore RR(E) \geq RR(C)$. So E executes then C executes.

(*) 2021 - Fall

Process	AT	BT
P ₁	0.0	7
P ₂	3.0	4
P ₃	5.0	2
P ₄	6.0	4

* Gantt chart:



$$RR(P) = \frac{\text{Time Now} - AT + BT}{BT}$$

At time = 7:

$$RR(P_2) = \frac{7 - 3 + 4}{4} = 2$$

$$RR(P_3) = \frac{7 - 5 + 2}{2} = 2$$

$$RR(P_4) = \frac{7 - 6 + 4}{4} = 1.25$$

$\therefore RR(P_2) = RR(P_3) > RR(P_4)$. P_2 arrives at first, so P_2 executes.

At time = 11

$$RR(P_3) = \frac{11 - 5 + 2}{2} = 4$$

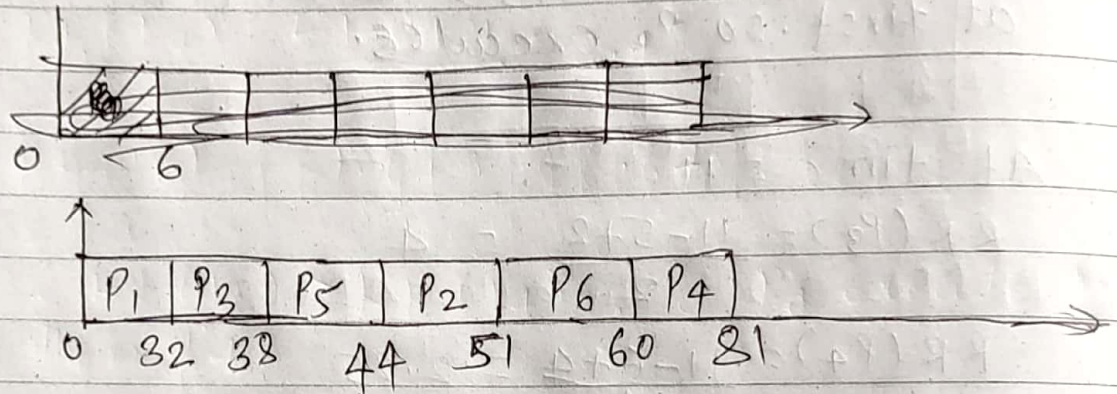
$$RR(P_4) = \frac{11 - 6 + 4}{4} = 2.25$$

$\therefore RR(P_3) > RR(P_4)$. so P_3 executes then P_4 executes.

2022-Fall

Process	BT	AT
P ₁	32	0
P ₂	7	0
P ₃	6	0
P ₄	21	0
P ₅	6	0
P ₆	9	0

* Gantt chart:



$$RR(P_i) = \frac{\text{Time now} - AT + BT}{BT}$$

At time = 0:

$$RR(P_1) = \frac{0 - 0 + 32}{32} = 1$$

$$RR(P_2) = \frac{0 - 0 + 7}{7} = 1$$

$$RR(P_3) = \frac{0 - 0 + 6}{6} = 1$$

$$RR(P_4) = \frac{0 - 0 + 21}{21} = 1$$

$$RR(P_5) = \frac{0 - 0 + 6}{6} = 1$$

$$RR(P_6) = \frac{0-0+9}{9} = 1$$

Since $RR(P_1) = RR(P_2) = RR(P_3) = RR(P_4) = RR(P_5) = RR(P_6)$, So P_1 executes first.

At time = 32:

$$RR(P_2) = \frac{32-0+7}{7} = 5.57$$

$$RR(P_3) = \frac{32-0+6}{6} = 6.33$$

$$RR(P_4) = \frac{32-0+21}{21} = 2.52$$

$$RR(P_5) = \frac{32-0+6}{6} = 6.33$$

$$RR(P_6) = \frac{32-0+9}{9} = 4.56$$

$\therefore RR(P_3) = RR(P_5) > RR(P_2) > RR(P_6) > RR(P_4)$. So P_3 executes.

At time = 38:

$$RR(P_2) = \frac{38-0+7}{7} = 6.4$$

$$RR(P_4) = \frac{38-0+21}{21} = 2.8$$

$$RR(P_5) = \frac{38-0+6}{6} = 7.33$$

$$RR(P_6) = \frac{38-0+9}{9} = 5.22$$

$\therefore RR(P_5) > RR(P_2) > RR(P_6) > RR(P_4)$. So P_5 executes.

At time = 44!

$$RR(P_2) = \frac{44 - 0 + 7}{7} = 7.28$$

$$RR(P_4) = \frac{44 - 0 + 21}{21} = 3.1$$

$$RR(P_6) = \frac{44 - 0 + 9}{9} = 5.89$$

$\therefore RR(P_2) > RR(P_6) > RR(P_4)$. So P_2 executes

At time = 51:

$$RR(P_4) = \frac{51 - 0 + 21}{21} = 3.43$$

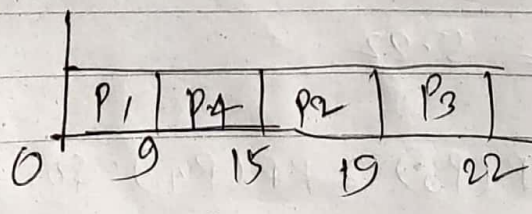
$$RR(P_6) = \frac{51 - 0 + 9}{9} = 6.67$$

$\therefore RR(P_6) > RR(P_4)$. So P_6 executes and then P_4 .

2023 - Spring

Process	AT	BT
P_1	0	9
P_2	4	4
P_3	10	3
P_4	0	6

* Gantt chart:



$$RR(P) = \frac{\text{Time} - AT + BT}{B}$$

At time 20:

$$RR(P_1) = \frac{0 - 0 + 9}{9} = 1$$

$$RR(P_2) = \frac{0 - 0 + 6}{6} = 1$$

$\therefore RR(P_1) = RR(P_2)$ ~~so~~ so P_1 executes.

At time ~~7~~ 9:

$$RR(P_2) = \frac{9 - 4 + 4}{4} = ~~2.25~~ 2.25$$

$$RR(P_4) = \frac{9 - 0 + 6}{6} = 2.5$$

~~RR(P_4)~~ $\therefore RR(P_4) > RR(P_2)$. So P_4 executes then P_2 .

At time = 15

$$RR(P_2) = \frac{15 - 4 + 4}{4} = 3.75$$

$$RR(P_3) = \frac{15 - 10 + 3}{3} = 2.67$$

$\therefore RR(P_2) > RR(P_3)$. So P_2 executes

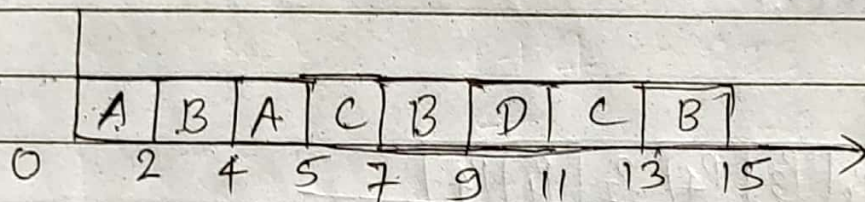
RR (Q = -)

Process	AT	BT
A	0	3 1 0 RR (Q = 2)
B	1	6 4 2 0
C	4	4 2 0
D	6	2 0

→

* Read Queue: ~~A~~~~B~~~~A~~~~C~~~~B~~~~D~~~~C~~~~B~~

* Gantt chart:



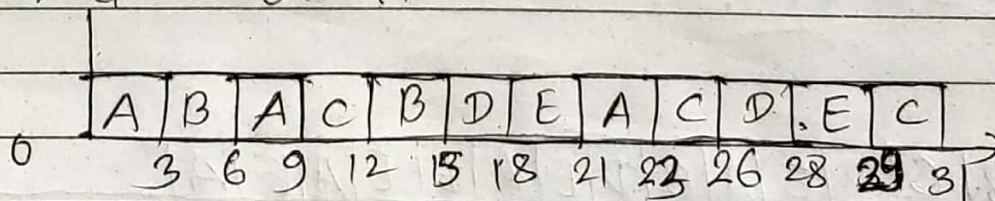
2020-Fall

RR (Q = 3)

Process	AT	BT
A	0	8 5 2 0
B	2	6 3 0
C	4	5 3 0
D	7	5 2 0
E	9	4 1 0

Ready Queue: ~~A~~~~B~~~~A~~~~C~~~~B~~~~D~~~~E~~~~A~~~~C~~~~D~~~~E~~~~C~~

* Gantt chart:



2021-Fall

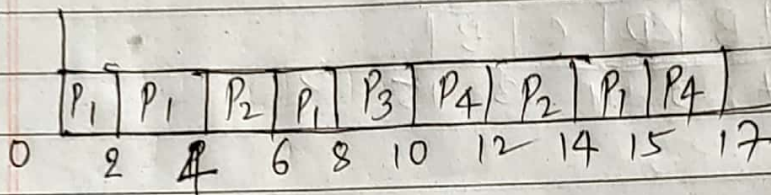
Date _____
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Q1 RR (Q=2):

Process	AT	BT
P ₁	0.0	7 8 3 1 0
P ₂	3.0	4 2 0
P ₃	5.0	2 0
P ₄	6.0	4 2 0

* Ready Queue: ~~P₁~~ ~~P₁~~ ~~P₂~~ ~~P₁~~ ~~P₃~~ ~~P₄~~ ~~P₂~~ ~~P₁~~ ~~P₄~~

* Gantt chart:

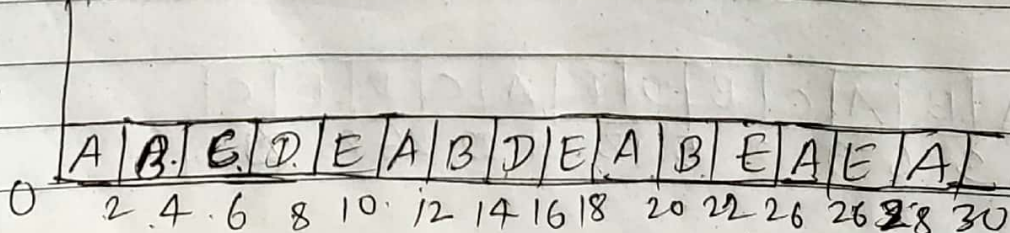


Q2 RR (Q=2):

Process	AT	BT
A	0	10 8 6 4 2 0
B	0	6 4 2 0
C	0	2 0
D	0	4 2 0
E	0	8 6 4 2 0

* Ready Queue: ~~A~~ ~~B~~ ~~C~~ ~~D~~ ~~E~~ ~~A~~ ~~B~~ ~~D~~ ~~E~~ ~~A~~ ~~B~~ ~~C~~ ~~A~~ ~~E~~ ~~A~~

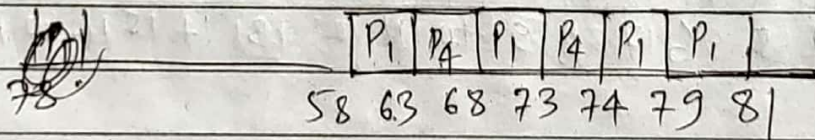
* Gantt chart:



RR (Q=5)

Ready Queue:

P_1	P_2	P_3	P_4	P_5	P_6	P_1	P_2	P_3	P_4	P_5	P_6	P_1	P_4	P_6	P_1	P_2	P_3	P_4
5	10	15	20	25	30	35	37	38	43	44	47	52	57	58	69	72	73	78



Q: R.R (Q=2)

Process AT BT

P₀ 0 3 1 0

P₁ 2 6 4 2 0

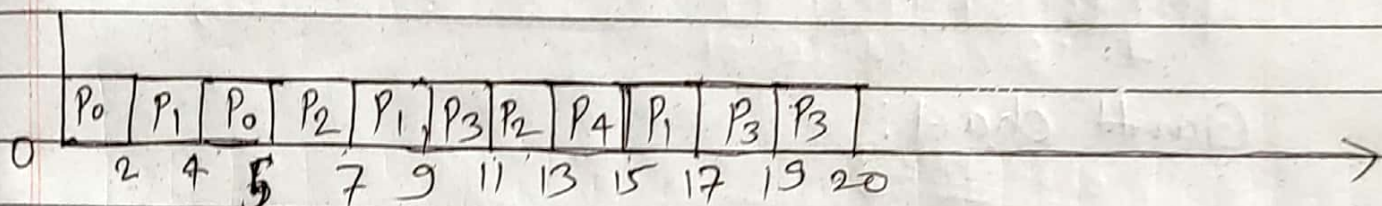
P₂ 4 4 2 0

P₃ 6 8 3 1

P₄ 8 2 0

Ready Queue: P₀ P₁ P₀ P₂ P₁ P₃ P₂ P₄ P₁ P₃ P₃

Gantt chart:



Q: R.R (Q=2)

Process AT BT

P₁ 0 4 2 0

P₂ 1 8 ~~* Ready Queue: 3 1 0~~

P₃ 2 2 0

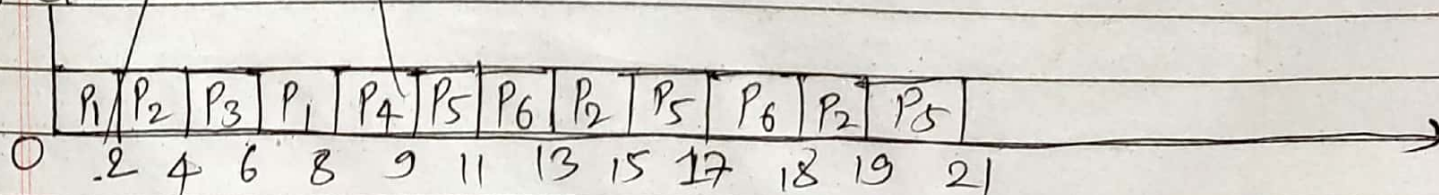
P₄ 3 1 0

P₅ 4 6 2 0

P₆ 6 1 3 1 0

R.Q = P₁ P₂ P₃ P₄ P₅ P₆ P₂ P₅ P₆ P₂ P₅

* Gantt chart:

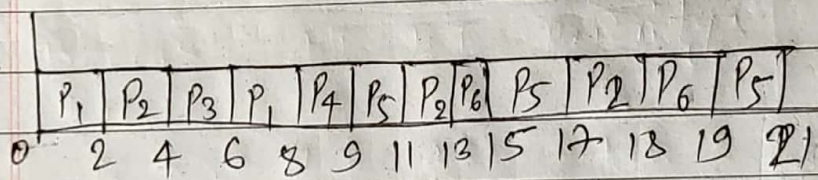


RR (Q=2)!

Process	AT	BT
P ₁	0	4 2 0
P ₂	1	5 3 1
P ₃	2	2 0
P ₄	3	1 0
P ₅	4	6 4 2 0
P ₆	6	3 1 0

Ready Queue: ~~P₁~~/~~P₂~~/~~P₃~~/~~P₁~~/~~P₄~~/~~P₅~~/~~P₂~~/~~P₆~~/~~P₅~~/~~P₂~~/~~P₆~~/~~P₅~~

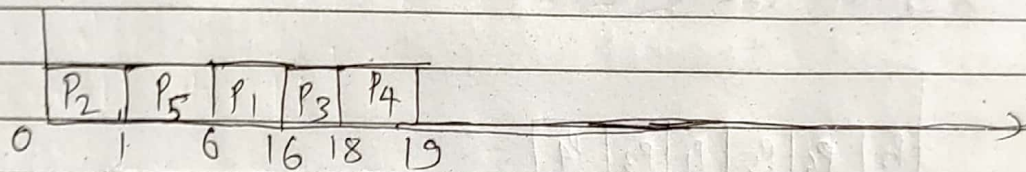
Gantt chart:



Priority Scheduling

Q	Process	BT	Priority	AT
	P ₁	10	3	0
✓	P ₂	1	1	0
	P ₃	2	4	0
✓	P ₄	1	5	0
	P ₅	5	2	0

• Gantt chart for lowest number has highest priority:

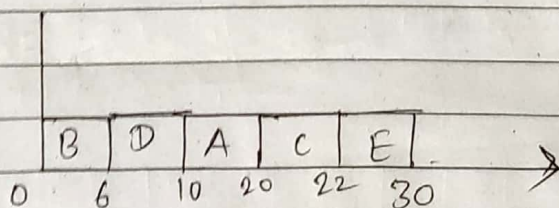


2021 ~~fall~~ Spring (↑ has ↑ priority)

Q	Process	AT	BT	Priority
	A	0	10	3
✓	B	0	6	5
	C	0	2	2
	D	0	4	4
	E	0	8	1

∵ 5 has ↑ priority

* Gantt chart:

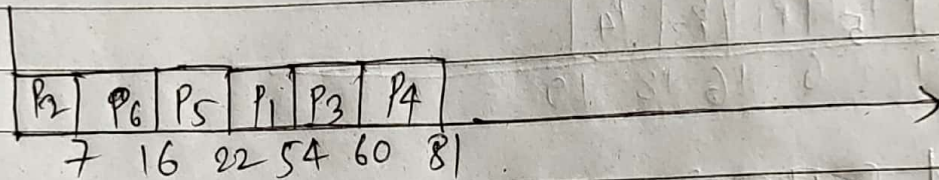


2022-Fall

Q1	Process	ATB	AT	priority
	P ₁	32	0	3
	P ₂	7	0	6
	P ₃	6	0	2
	P ₄	21	0	1
	P ₅	6	0	4
	P ₆	9	0	5

note:
(↑ has ↑ priority)

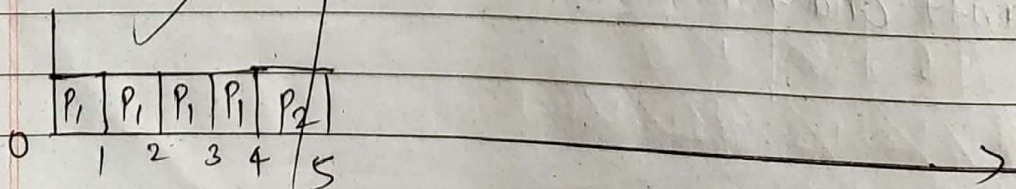
* Gantt chart



2023-Spring

Process	AT	BT	priority	TA
P ₁	0	9	2	84
P ₂	4	4	1	3
P ₃	10	3	3	0
P ₄	0	6	4	0

* Gantt chart:

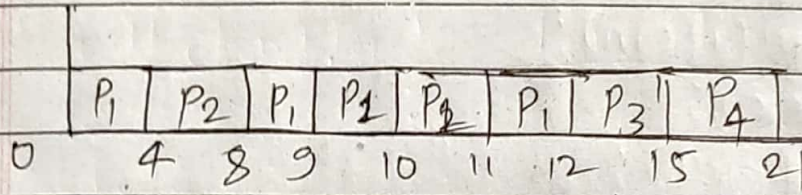


2023-spring

Date _____
Page _____

Process	AT	BT	Priority
P ₁	0	9	2
P ₂	4	4	1
P ₃	10	3	3
P ₄	0	6	4

* Gantt chart:



Notes

Process	BT	Priority
P ₁	10	3
P ₂	4	1
P ₃	2	4
P ₄	1	5
P ₅	8	2

(P₂ has ↑ priority)

* Gantt chart:

